AGN Final Project

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[410]:
```
# Libraries
import sys
sys.path.append('/Users/isaac/Documents/MASS/Academic/2nd semester/AGN/
↪tutorials/AGN-tutorials/SciScript-Python')

from SciServer import Authentication, CasJobs

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.patches as mpatches
from matplotlib.patches import Ellipse, FancyArrowPatch
plt.style.use('seaborn-v0_8-deep')
plt.rcParams.update({'font.size': 12})

from dotenv import load_dotenv
import os
```

1 Optical Spectral Classification (BPT & WHAN)

1.1 Sample Selection

The selection is designed to isolate narrow-line emission galaxies with reliable line measurements:

- **FWHM < 500 km/s** on the Balmer and [O III] lines removes Type 1 AGN, whose broad permitted lines originate in the high-velocity broad-line region (BLR). We want only narrow-line objects so the diagnostics trace the NLR / host gas.
- **SNR > 3** on every diagnostic line ensures the ratios are not dominated by noise.
- **Emission requirement** (equivalent width < -0.1 Å, negative = emission) on all lines

Additionally, we restrict the **median SNR in g band > 20** to improve the reliability on the [OIII] velocity dispersion calculations

[411]:
```
# Run query and save results to CSV

# SciServer authentication
load_dotenv()
username = os.getenv("SCISERVER_USER")
```

```python
password = os.getenv("SCISERVER_PASS")

Authentication.login(username, password)

# I include from the start the wise query to use in next tasks
query = '''
    SELECT TOP 10000
        s.specobjid,s.z,
        g.oiii_5007_flux,
        g.oiii_sigma,
        g.oi_6300_flux,
        g.h_beta_flux,
        g.h_alpha_flux,
        g.h_alpha_eqw,
        g.sii_6717_flux,
        g.sii_6731_flux,
        g.nii_6584_flux,
        g.oiii_5007_flux / g.h_beta_flux   AS oiii_hbeta_ratio,
        g.nii_6584_flux  / g.h_alpha_flux  AS nii_halpha_ratio,
        w.w1mpro AS w1,
        w.w2mpro AS w2,
        w.w3mpro AS w3,
        w.w1mpro - w.w2mpro AS w1_w2,
        w.w2mpro - w.w3mpro AS w2_w3

    FROM SpecObj AS s
        JOIN GalSpecLine AS g ON s.specobjid = g.specobjid
        JOIN photoTag    AS p ON s.bestobjid = p.objid
        LEFT JOIN wise_xmatch AS x ON p.objid = x.sdss_objid --- The LEFT JOIN ensures we keep all SDSS objects, even those without a WISE match so that we can count them later
                                AND x.match_dist < 3.0 --- Only consider WISE matches within 3 arcseconds
                                AND x.match_dist = (SELECT MIN(match_dist)
                                                    FROM wise_xmatch
                                                    WHERE sdss_objid = p.objid) --- Select the closest WISE match for each SDSS object
        LEFT JOIN wise_allsky AS w ON x.wise_cntr = w.cntr

    WHERE
        (s.class = 'GALAXY' OR s.class = 'QSO') AND
        s.z          < 0.35     AND
        s.snmedian_g > 20       AND

        -- Velocity dispersion (FWHM < 500 km/s)
        2.3548 * g.sigma_balmer < 500 AND
        2.3548 * g.oiii_sigma   < 500 AND
```

```
        -- Emission line detection (negative EQW = emission)
        g.oiii_5007_eqw  < -0.1 AND
        g.oi_6300_eqw    < -0.1 AND
        g.h_beta_eqw     < -0.1 AND
        g.h_alpha_eqw    < -0.1 AND
        g.nii_6584_eqw   < -0.1 AND
        g.sii_6717_eqw   < -0.1 AND
        g.sii_6731_eqw   < -0.1 AND

        -- Minimum absolute flux
        g.oiii_5007_flux > 5 AND
        g.oi_6300_flux   > 5 AND
        g.h_beta_flux    > 5 AND
        g.h_alpha_flux   > 5 AND
        g.nii_6584_flux  > 5 AND
        g.sii_6717_flux  > 5 AND
        g.sii_6731_flux  > 5 AND

        -- flux > 5 * flux_err for all diagnostic lines
        g.oiii_5007_flux > 5 * g.oiii_5007_flux_err AND
        g.oi_6300_flux   > 5 * g.oi_6300_flux_err   AND
        g.h_beta_flux    > 5 * g.h_beta_flux_err     AND
        g.h_alpha_flux   > 5 * g.h_alpha_flux_err    AND
        g.nii_6584_flux  > 5 * g.nii_6584_flux_err  AND
        g.sii_6717_flux  > 5 * g.sii_6717_flux_err  AND
        g.sii_6731_flux  > 5 * g.sii_6731_flux_err

        ORDER by s.specobjid -- We order by specobjid to ensure that the same␣
 ↪objects are returned in the same order if we run the query multiple times
'''

#result = CasJobs.executeQuery(query, context='DR18')
#result.to_csv('agn_sample.csv', index=False)
result = pd.read_csv('agn_sample.csv')
```

[412]:
```
# Calculate how many SDSS objects have a WISE match (non-NaN w1)
total        = len(result)
wise_matched = result['w1'].notna().sum()
no_match     = total - wise_matched
duplicates   = result['specobjid'].duplicated().sum()  # Check for duplicate␣
 ↪specobjid entries

# Print results
print(f"SDSS objects:        {total}")
print(f"With WISE match:     {wise_matched}")
print(f"No WISE counterpart: {no_match}")
```

```
print(f"Duplicate entries:   {duplicates}")
```

```
SDSS objects:         10000
With WISE match:      9854
No WISE counterpart: 146
Duplicate entries:    0
```

[413]:
```
# Do all the next steps with objects that have WISE data
#data = result[result['w1'].notna()]
#data.to_csv('agn_sample_wise.csv', index=False)
```

[414]:
```
data = pd.read_csv('agn_sample_wise.csv')
print(data.columns)
```

```
Index(['specobjid', 'z', 'oiii_5007_flux', 'oiii_sigma', 'oi_6300_flux',
       'h_beta_flux', 'h_alpha_flux', 'h_alpha_eqw', 'sii_6717_flux',
       'sii_6731_flux', 'nii_6584_flux', 'oiii_hbeta_ratio',
       'nii_halpha_ratio', 'w1', 'w2', 'w3', 'w1_w2', 'w2_w3'],
      dtype='str')
```

1.2 BPT

Separation lines

Theoretical SF vs AGN division (Kewley et al. 2001). Places an upper limit on the maximum line ratios achievable by pure photoionization from stars:

$$\log([\mathrm{OIII}]/\mathrm{H}\beta) = \frac{0.61}{\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.47} + 1.19$$

Empirical SF vs AGN division (Kaufmann et al. 2003). Revised division between SF galaxies and AGN to account for the scatter due correlattions between properties such as ionization parameter and metallicity:

$$\log([\mathrm{OIII}]/\mathrm{H}\beta) = \frac{0.61}{\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.05} + 1.3$$

Emprical Seyfert-LINER division (Schawinski et. al 2007)

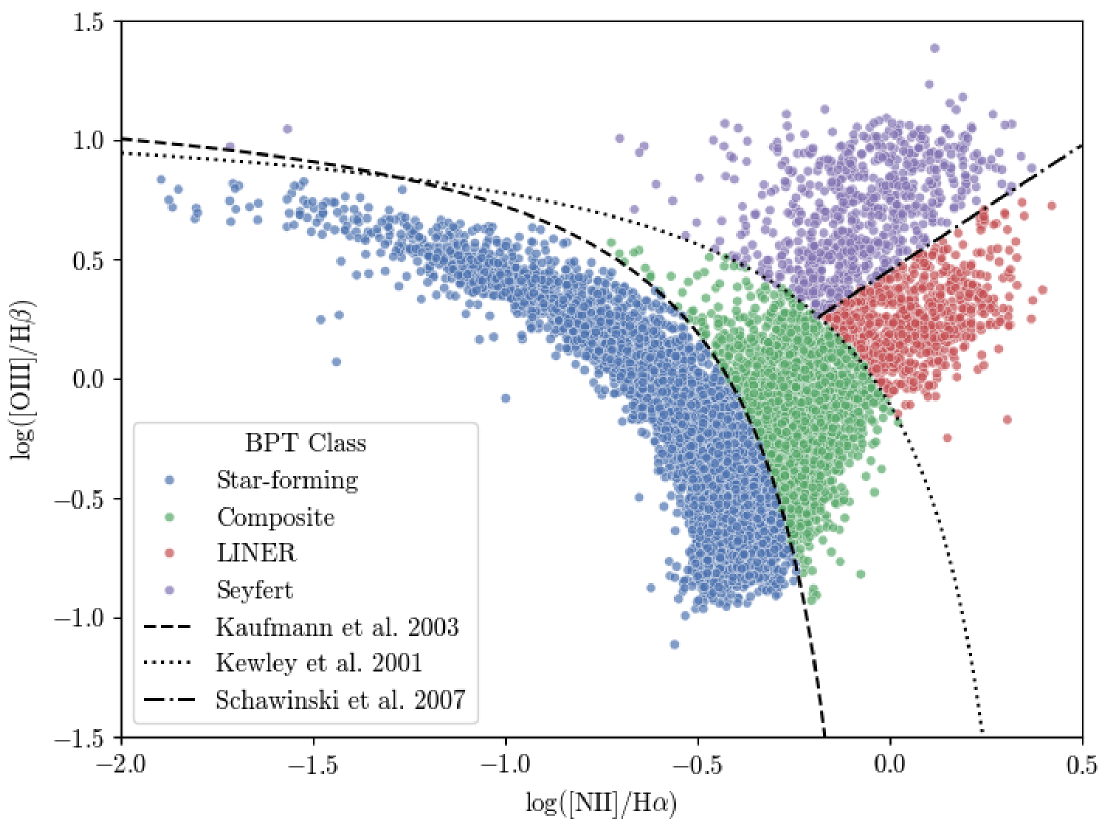
$$\log([\mathrm{OIII}]/\mathrm{H}\beta) = 1.05 \log(\mathrm{NII}]/\mathrm{H}\alpha) + 0.45$$

The *composite* objects are in the transition between Kewley and Kauffman lines

[415]:
```
# Classification lines
log_x_kauf  = np.linspace(-2, 0.05, 200)
log_x_kew   = np.linspace(-2, 0.47, 200)
log_x_schaw = np.linspace(np.log10(0.65), 0.5, 200)

log_x_kauf  = log_x_kauf[log_x_kauf < 0.05]
```

```
log_x_kew   = log_x_kew[log_x_kew < 0.47]
log_x_schaw = log_x_schaw[log_x_schaw > np.log10(0.65)]

def kauffmann(log_nii_halpha):
    return 0.61 / (log_nii_halpha - 0.05) + 1.3

def kewley(log_nii_halpha):
    return 0.61 / (log_nii_halpha - 0.47) + 1.19

def schawinski(log_nii_halpha):
    return 1.05 * log_nii_halpha + 0.45

# Classify objects
log_nii = np.log10(data['nii_halpha_ratio'])
log_oiii = np.log10(data['oiii_hbeta_ratio'])

data['log_nii'] = log_nii
data['log_oiii'] = log_oiii

below_kauffmann = (log_nii < 0.05) & (log_oiii < kauffmann(log_nii))
above_kewley    = (log_nii < 0.47) & (log_oiii > kewley(log_nii)) | (log_nii >=␣
↪0.47)
above_schawinski = log_oiii > schawinski(log_nii)

sf         = below_kauffmann
agn        = above_kewley & ~sf
composite = ~sf & ~agn
seyfert   = above_kewley & above_schawinski
liners    = above_kewley & ~above_schawinski

data['BPT class'] = 'Unclassified'
data.loc[sf, 'BPT class'] = 'Star-forming'
data.loc[agn, 'BPT class'] = 'AGN'
data.loc[composite, 'BPT class'] = 'Composite'
data.loc[seyfert, 'BPT class'] = 'Seyfert'
data.loc[liners, 'BPT class'] = 'LINER'

# Plot
plt.figure(figsize=(8,6))

# Data points
sns.scatterplot(data,x='log_nii',y='log_oiii',hue='BPT class',s=20,alpha=0.7)

# Classification lines
plt.plot(log_x_kauf, 0.61 / (log_x_kauf - 0.05) + 1.3, color='k',␣
↪linestyle='--', label='Kaufmann et al. 2003')
```

```
plt.plot(log_x_kew, 0.61 / (log_x_kew - 0.47) + 1.19, color='k', linestyle=':',␣
 ↪label='Kewley et al. 2001')
plt.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45, color='k', linestyle='-.',␣
 ↪label='Schawinski et al. 2007')

# Plot settings
plt.xlabel(r'$\log([\mathrm{NII}] / \mathrm{H}\alpha)$')
plt.ylabel(r'$\log([\mathrm{OIII}] / \mathrm{H}\beta)$')
plt.xlim(-2, 0.5)
plt.ylim(-1.5, 1.5)
plt.legend(title='BPT Class')
plt.show()
```



[416]:
```
print('----- AGN BPT Classification -----\n')
print('Objects Classified as Strong AGN (Seyfert):\n',
      f'Number: {(data["BPT class"] == "Seyfert").sum()}\n',
      f'Percentage: {(data["BPT class"] == "Seyfert").sum() / len(data) * 100:.
 ↪2f}%')

print('\nObjects Classified as LINERs:\n',
```

```
      f'Number: {(data["BPT class"] == "LINER").sum()}\n',
      f'Percentage: {(data["BPT class"] == "LINER").sum() / len(data) * 100:.
 ↪2f}%')

print('\nTotal objects classified as AGN (Seyfert + LINER):\n',
      f'Number: {(data["BPT class"].isin(["Seyfert", "LINER"])).sum()}\n',
      f'Percentage: {(data["BPT class"].isin(["Seyfert", "LINER"])).sum() /␣
 ↪len(data) * 100:.2f}%')
```

```
----- AGN BPT Classification -----

Objects Classified as Strong AGN (Seyfert):
 Number: 729
 Percentage: 7.40%

Objects Classified as LINERs:
 Number: 725
 Percentage: 7.36%

Total objects classified as AGN (Seyfert + LINER):
 Number: 1454
 Percentage: 14.76%
```

1.3 WHAN

$W_{\mathrm{H}\alpha} = 6$ is the transposed version of Kewley et al 2006 Seyfert/LINER division (which uses [SII]/Hα and [OI]/Hα diagrams)

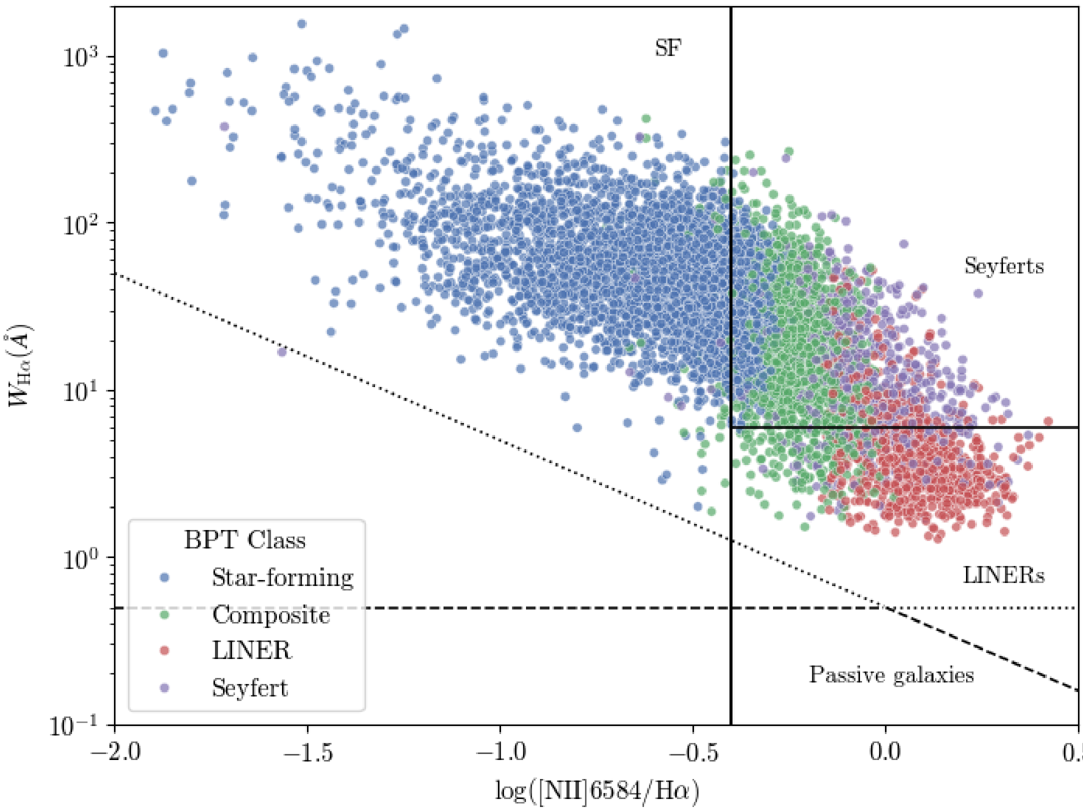
$\log[\mathrm{NII/H}\alpha] = -0.4$ is the SF/AGN division

Dotted lines mark $W_{\mathrm{H}\alpha} = 0.5$ and $W_{[\mathrm{NII}]} = 0.5$ limits, below which line measurements are uncertain.

In this case we don't see passive galaxies ($W_{\mathrm{H}\alpha}$ and $W_{[\mathrm{NII}]} < 0.5$).

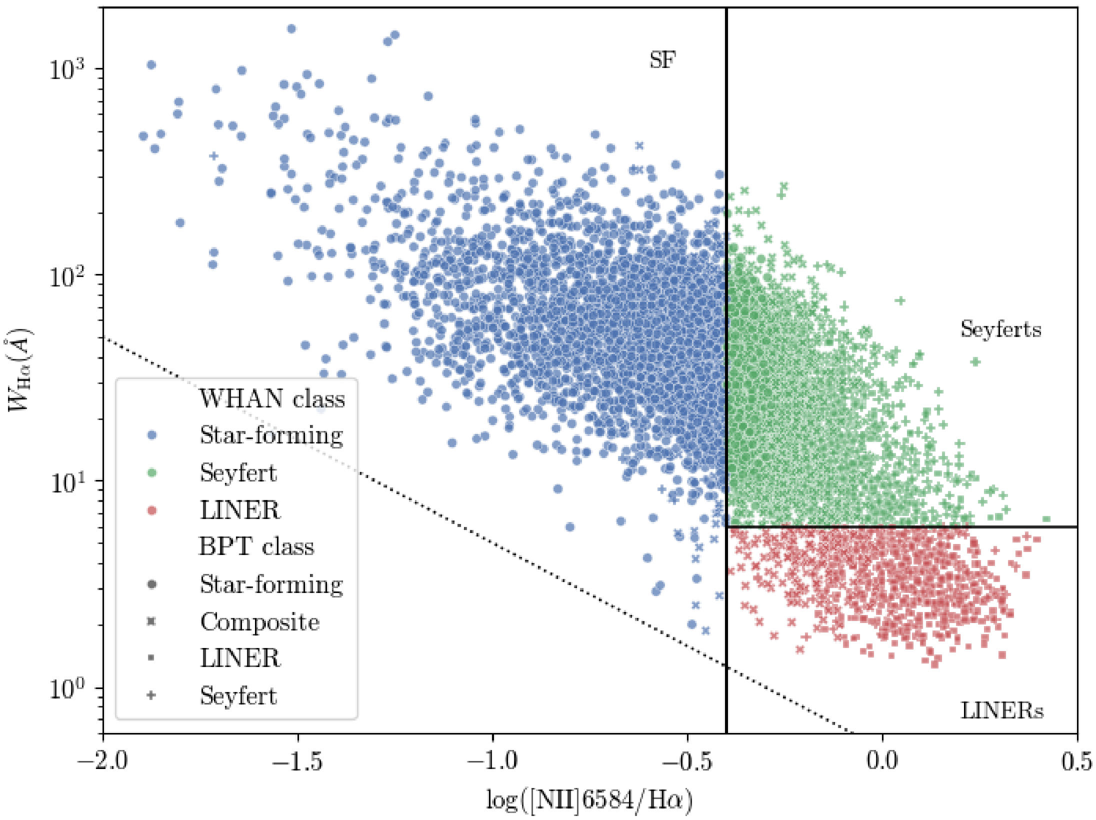
[417]:
```
# Plot WHAN diagram with BPT classification
plt.figure(figsize=(8,6))

data['abs_h_alpha_eqw'] = np.abs(data['h_alpha_eqw'])
sns.scatterplot(data,y='abs_h_alpha_eqw',x='log_nii',hue='BPT␣
 ↪class',s=20,alpha=0.7)

xx = np.linspace(-2, 0.6, 500)

# Classification lines
plt.axvline(x=-0.4, color='k') # SF vs AGN
plt.plot(xx[xx>-0.4], 6*np.ones_like(xx[xx>-0.4]), color='k', linewidth=1.2) #␣
 ↪Seyfert vs LINER

# Passive galaxies vs AGN
```

```python
plt.plot(xx[xx<0], 0.5*np.ones_like(xx[xx<0]), color='k', linestyle='--',␣
↪linewidth=1.2)
plt.plot(xx[xx>=0], 0.5*np.ones_like(xx[xx>=0]), color='k', linestyle=':',␣
↪linewidth=1.2)

# Uncertain measurements (W_Ha and W_NII < 0.5 A)
yy = 0.5/(10**xx)
plt.plot(xx[xx<0], yy[xx<0], color='k', linestyle=':', linewidth=1.2)
plt.plot(xx[xx>=0], yy[xx>=0], color='k', linestyle='--', linewidth=1.2)

# Region labels
plt.text(-0.6,  1000, 'SF', fontsize=11)
plt.text(0.2,  50, 'Seyferts', fontsize=11)
plt.text(0.2,  0.7, 'LINERs', fontsize=11)
plt.text(-0.2,  0.18, 'Passive galaxies',fontsize=11)

# Plot settings
plt.ylabel(r'$W_{\mathrm{H}\alpha}(\AA)$')
plt.xlabel(r'$\log([\mathrm{NII}] 6584 / \mathrm{H}\alpha)$')
plt.yscale('log')
x_lim = (-2, 0.5)
y_lim = (0.1, 2000)
plt.xlim(x_lim)
plt.ylim(y_lim)
plt.legend(loc='lower left',title='BPT Class')
plt.show()
```



[418]:
```
# Define WHAN class and plot
data['WHAN class'] = 'Unclassified'
data.loc[(data['nii_halpha_ratio'] > 0.4 ) & (data['abs_h_alpha_eqw'] > 6),␣
↪'WHAN class'] = 'Seyfert'
data.loc[(data['nii_halpha_ratio'] > 0.4 ) & (data['abs_h_alpha_eqw'] <= 6),␣
↪'WHAN class'] = 'LINER'
data.loc[(data['nii_halpha_ratio'] <= 0.4) & (data['abs_h_alpha_eqw'] > 0.5),␣
↪'WHAN class'] = 'Star-forming'

# Count objects classified as Seyfert in WHAN diagram that also are Seyfert in␣
↪BPT diagram
seyfert_whan = (data['WHAN class'] == 'Seyfert')
both_seyfert = seyfert_whan & (data['BPT class'] == 'Seyfert')

plt.figure(figsize=(8,6))

sns.scatterplot(data,y='abs_h_alpha_eqw',x='log_nii',hue='WHAN␣
↪class',style='BPT class',s=20,alpha=0.7)

xx = np.linspace(-2, 0.6, 500)
```

```
# Classification lines
plt.axvline(x=-0.4, color='k') # SF vs AGN
plt.plot(xx[xx>-0.4], 6*np.ones_like(xx[xx>-0.4]), color='k', linewidth=1.2) #␣
 ↪Seyfert vs LINER

# Passive galaxies vs AGN
plt.plot(xx[xx<0], 0.5*np.ones_like(xx[xx<0]), color='k', linestyle='--',␣
 ↪linewidth=1.2)
plt.plot(xx[xx>=0], 0.5*np.ones_like(xx[xx>=0]), color='k', linestyle=':',␣
 ↪linewidth=1.2)

# Uncertain measuremens (W_Ha and W_NII < 0.5 A)
yy = 0.5/(10**xx)
plt.plot(xx[xx<0], yy[xx<0], color='k', linestyle=':', linewidth=1.2)
plt.plot(xx[xx>=0], yy[xx>=0], color='k', linestyle='--', linewidth=1.2)

# Region labels
plt.text(-0.6,  1000, 'SF', fontsize=11)
plt.text(0.2,  50, 'Seyferts', fontsize=11)
plt.text(0.2,  0.7, 'LINERs', fontsize=11)
#plt.text(-0.2,  0.18, 'Passive galaxies',fontsize=11)

# Plot settings
plt.ylabel(r'$W_{\mathrm{H}\alpha}(\AA)$')
plt.xlabel(r'$\log([\mathrm{NII}] 6584 / \mathrm{H}\alpha)$')
plt.yscale('log')
x_lim = (-2, 0.5)
y_lim = (0.6, 2e3)
plt.xlim(x_lim)
plt.ylim(y_lim)
plt.legend(loc='lower left')
plt.show()
```



[419]:
```
print('----- WHAN vs BPT Comparison -----\n')

print('Objects classified as Seyfert in WHAN diagram:\n'
      f'Number: {seyfert_whan.sum()}\n'
      f'Percentage: {seyfert_whan.sum() / len(data) * 100:.2f}%\n')

print('Objects classified as Seyfert in both WHAN and BPT diagram:\n'
      f'Number: {both_seyfert.sum()}\n'
      f'Percentage: {both_seyfert.sum() / len(data) * 100:.2f}%')
```

```
----- WHAN vs BPT Comparison -----

Objects classified as Seyfert in WHAN diagram:
Number: 3888
Percentage: 39.46%

Objects classified as Seyfert in both WHAN and BPT diagram:
Number: 573
Percentage: 5.81%
```

We can see that there is a big discrepancy between the BPT and WHAN classification. This could be due to the fact that the Seyfert WHAN region absorbs genuine weak AGN and some composites that the BPT counts in separate classes.

2 Infrared Classification (WISE)

What WISE colors trace

A luminous AGN heats dust near the nucleus, which radiates a strong mid-infrared continuum that makes the AGN appear *red* in $W1 - W2$. Star-forming and galaxies have cooler dust and sit *bluer*

Mateos et al 2012 wedge

In Vega magnitudes (as in the WISE catalogue), the AGN locus is

$$y = 0.315x$$

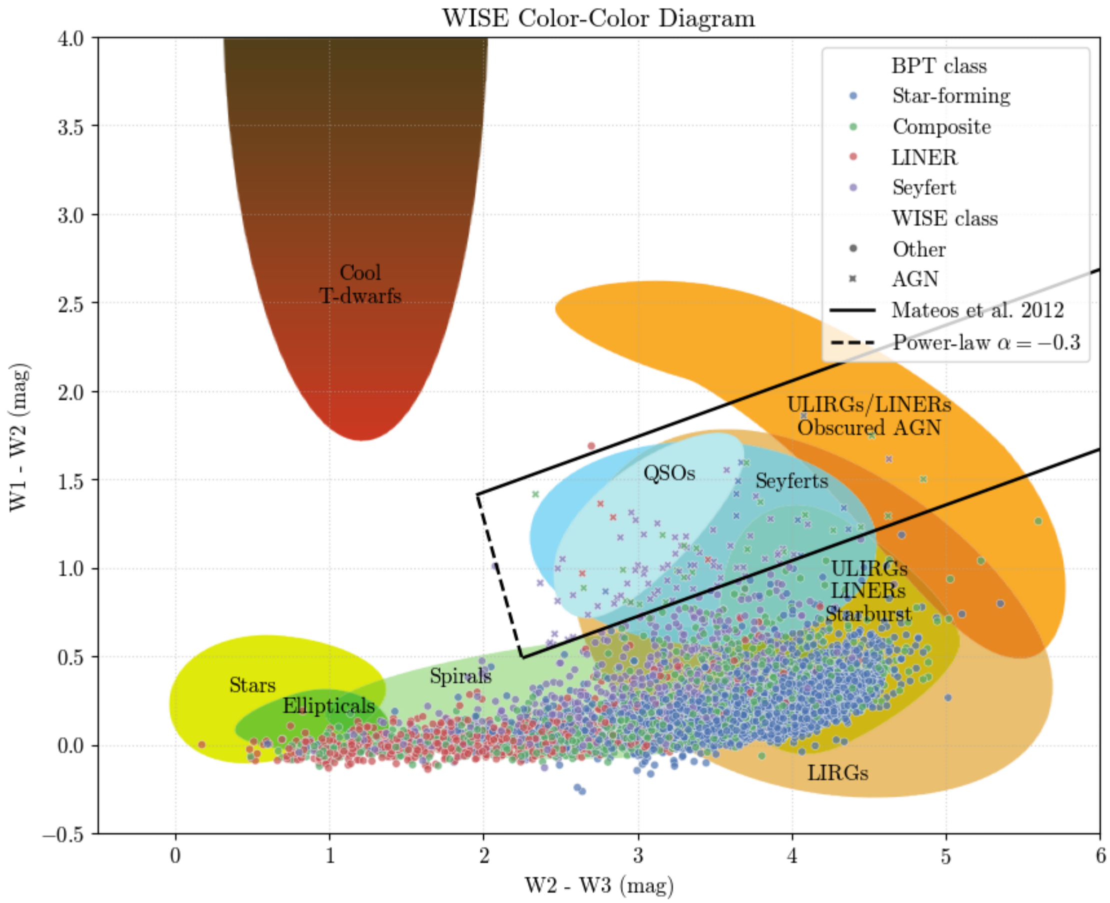
where $x \equiv w_2 - w_3$ and $y \equiv w_1 - 2_2$. The top and and bottom boundaries are obtained by adding y-axis intercepts of $+0.796$ and -0.222, respectively. The MIR powerlaw $\alpha = -0.3$ bottom-left limit corresponds to

$$y = -3.172x + 7.624$$

[420]:
```python
# Mateos et al 2012 wedge boundaries (converted to magnitude space)

x = data['w2_w3']    # W2 - W3 [Vega mag]
y = data['w1_w2']    # W1 - W2 [Vega mag]

def top_boundary(x):    return  0.315 * x + 0.796
def bottom_boundary(x): return  0.315 * x - 0.220
def powerlaw_limit(x):  return -3.172 * x + 7.624

# Wedge selection mask
in_wedge = (
    (y <  top_boundary(x))    &    # below top line
    (y >  bottom_boundary(x)) &    # above bottom line
    (y >  powerlaw_limit(x))       # above power-law bottom-left limit
)

# Add WISE class based on wedge
data['WISE class'] = 'Other'
data.loc[in_wedge, 'WISE class'] = 'AGN'
```

[421]:
```python
fig, ax = plt.subplots(figsize=(10,8))

import matplotlib.image as mpimg

img = mpimg.imread('blobs.tiff')

plt.imshow(img,extent=[-1, 6, -0.5, 4], aspect='auto')
```

```python
ax.text(1.2, 2.5, 'Cool\nT-dwarfs', color='k',ha='center')
ax.text(0.50, 0.30, 'Stars', color='k', ha='center')
ax.text(1.85, 0.35, 'Spirals', color='k', ha='center')
ax.text(1.0, 0.18, 'Ellipticals', color='k', ha='center')
ax.text(4, 1.45, 'Seyferts',color='k', ha='center')
ax.text(3.2, 1.5, 'QSOs', color='k', ha='center')
ax.text(4.5, 1.75, 'ULIRGs/LINERs\nObscured AGN',color='k', ha='center')
ax.text(4.5, 0.7, 'ULIRGs\nLINERs\nStarburst',color='k', ha='center')
ax.text(4.30, -0.20, 'LIRGs', color='k', ha='center')

# Objects from our sample with BPT classification
sns.scatterplot(data=data, x='w2_w3', y='w1_w2',hue='BPT class',style='WISE class', s=20,alpha=0.7)

# Matteos et al 2012 Wedge boundaries
x_range = np.linspace(-0.5, 6, 500)
ax.plot(x_range[x_range>1.95], top_boundary(x_range[x_range>1.95]),    'k', lw=1.8, label='Mateos et al. 2012')
ax.plot(x_range[x_range>2.26], bottom_boundary(x_range[x_range>2.26]), 'k', lw=1.8)
ax.plot(x_range[(x_range>1.96) & (x_range<2.26)], powerlaw_limit(x_range[(x_range>1.96) & (x_range<2.26)]),  'k--',  lw=1.8, label=r'Power-law $\alpha = -0.3$')

# Plot settings
ax.set_xlim(-0.5, 6)
#ax.set_ylim(-0.5,2)
ax.set_ylabel('W1 - W2 (mag)')
ax.set_xlabel('W2 - W3 (mag)')
ax.set_title('WISE Color-Color Diagram')
plt.legend(loc='upper right', bbox_to_anchor=(1, 1))
plt.grid(True, linestyle=':', alpha=0.5)
plt.show()
```



[422]:
```
# Count objects inside the Mateos wedge that are also classified as AGN␣
↪(Seyfert) in BPT diagram
wedge_agn_bpt = in_wedge & (data['BPT class'] == 'Seyfert')

# Count objects inside the Mateos wedge that are also classified as AGN␣
↪(Seyfert) in WHAN diagram
wedge_agn_whan = in_wedge & (data['WHAN class'] == 'Seyfert')
```

[423]:
```
print('----- WISE Color-Color Diagram Analysis -----\n')

print("Objects inside Mateos wedge (Classified as AGN by WISE):\n"
      f'Number: {in_wedge.sum()}\n'
      f'Percentage: {in_wedge.sum() / len(data) * 100:.2f}%')

print("\nObjects inside Mateos wedge that are also Strong AGN (Seyfert) in BPT␣
↪diagram:\n"
      f'Number: {wedge_agn_bpt.sum() }\n'
      f'Percentage: {wedge_agn_bpt.sum() / len(data) * 100:.2f}%')
```

```
print("\nObjects inside Mateos wedge that are also Strong AGN (Seyfert) in the␣
 ↪WHAN diagram:\n"
      f'Number: {wedge_agn_whan.sum() }\n'
      f'Percentage: {wedge_agn_whan.sum() / len(data) * 100:.2f}%')
```

```
----- WISE Color-Color Diagram Analysis -----

Objects inside Mateos wedge (Classified as AGN by WISE):
Number: 106
Percentage: 1.08%

Objects inside Mateos wedge that are also Strong AGN (Seyfert) in BPT diagram:
Number: 68
Percentage: 0.69%

Objects inside Mateos wedge that are also Strong AGN (Seyfert) in the WHAN
diagram:
Number: 82
Percentage: 0.83%
```

Discrepancy between WISE and optical AGN selection

- WISE selection is biased toward luminous, IR-bright AGN
- It misses low-luminosity Seyferts and LINERs

[424]:
```
# Objects that are AGN in all surveys
wedge_agn_all = in_wedge & (data['BPT class'] == 'Seyfert') & (data['WHAN␣
 ↪class'] == 'Seyfert')

print("\nObjects classified as AGN in WISE, BPT, and WHAN diagrams:\n"
      f'Number: {wedge_agn_all.sum()}\n'
        f'Percentage: {wedge_agn_all.sum() / len(data) * 100:.2f}%')
```

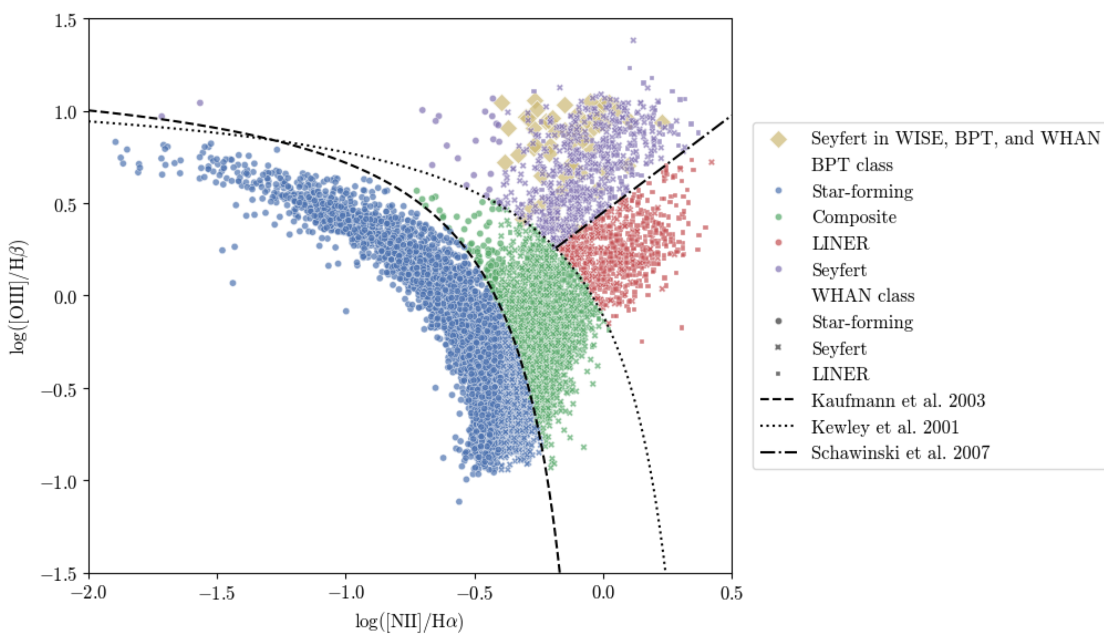
```
Objects classified as AGN in WISE, BPT, and WHAN diagrams:
Number: 58
Percentage: 0.59%
```

Summary of AGN Classification with SDSS and WISE

- From **10000** objects, **9854** had a WISE counterpart. From these, **1454** were classified as AGN from the Kewly and Kaufmann lines. From this **729** were classified as Strong AGN (Seyferts), which represent *7.4%* of the total.
- Then from the WHAN diagram **3888** objects are classified as Seyferts and **573** are Seyferts in both BPT and WHAN diagrams. This corresponds to *5.81%* of the total.
- Next, using the WISE color-color digram and the Mateos et al wedge, **106** objects are classified as AGN. Only **68** of these are also strong AGN (Seyferts) in the BPT diagram (*0.69%*) and **82** are Seyfert in the WHAN diagram (*0.83%*)
- Finally, **58** are classified as Seyferts in all surveys *(0.59%)*.

[425]:

```
# Plot the BPT diagram again, highlighting objects that are AGN in all three␣
↪classifications
data['AGN in all'] = wedge_agn_all

plt.figure(figsize=(7.5,6.5))
# Data points
sns.scatterplot(data[wedge_agn_all],x='log_nii',y='log_oiii',s=70,alpha=0.
↪7,marker='D', label='Seyfert in WISE, BPT, and WHAN',color='C4')
sns.scatterplot(data[~wedge_agn_all],x='log_nii',y='log_oiii',hue='BPT␣
↪class',style='WHAN class',s=20,alpha=0.7)

# Classification lines
plt.plot(log_x_kauf, 0.61 / (log_x_kauf - 0.05) + 1.3, color='k',␣
↪linestyle='--', label='Kaufmann et al. 2003')
plt.plot(log_x_kew, 0.61 / (log_x_kew - 0.47) + 1.19, color='k', linestyle=':',␣
↪label='Kewley et al. 2001')
plt.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45, color='k', linestyle='-.',␣
↪label='Schawinski et al. 2007')

# Plot settings
plt.xlabel(r'$\log([\mathrm{NII}] / \mathrm{H}\alpha)$')
plt.ylabel(r'$\log([\mathrm{OIII}] / \mathrm{H}\beta)$')
plt.xlim(-2, 0.5)
plt.ylim(-1.5, 1.5)
plt.legend(loc='center right', bbox_to_anchor=(1.6, 0.5))
plt.show()
```



3 NLR Gas Properties & SMBH Masses

3.1 Electron density

(Sanders et al 2015)

$$R(n_e) = a\frac{b + n_e}{c + n_e}$$

where $R = \text{[SII]}\lambda6716/\lambda6731, a = 0.4315, b = 2107, c = 627.1$

The two lines here arise from levels with nearly identical excitation energies but **different critical densities**, so their ratio is essentially independent of temperature and depends almost entirely on electron density through collisional de-excitation.

[426]:

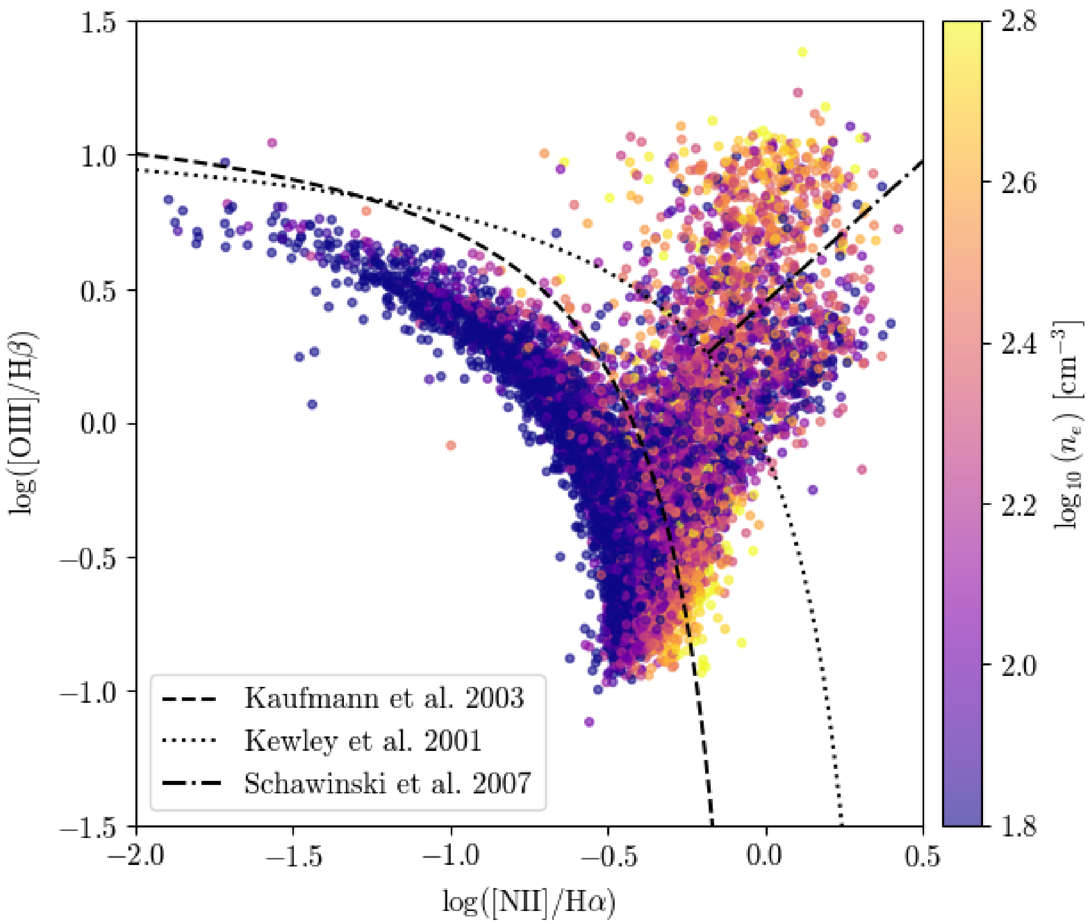
```python
# Calculate electron density from [SII] ratio
def ne(R_sii):
    a = 0.4315
    b = 2107.0
    c = 627.1
    return (c*R_sii - a*b) / (a - R_sii)

R_sii = data['sii_6717_flux'] / data['sii_6731_flux']

data['n_e'] = ne(R_sii)
```

[427]:

```python
# Plot BPT diagram colored by electron density
fig, ax = plt.subplots(figsize=(6.5, 5.5))

sc = ax.scatter(data['log_nii'], data['log_oiii'],
    c=np.log10(data['n_e']),
    cmap="plasma",
    vmin=1.8, vmax=2.8,
    s=10,
    alpha=0.6,
)

cbar = fig.colorbar(sc, ax=ax, pad=0.02)
cbar.set_label(r"$\log_{10}(n_e)$ [cm$^{-3}$]")

# Classification lines
plt.plot(log_x_kauf, 0.61 / (log_x_kauf - 0.05) + 1.3, color='k',␣
 ↪linestyle='--', label='Kaufmann et al. 2003')
plt.plot(log_x_kew, 0.61 / (log_x_kew - 0.47) + 1.19, color='k', linestyle=':',␣
 ↪label='Kewley et al. 2001')
plt.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45, color='k', linestyle='-.',␣
 ↪label='Schawinski et al. 2007')
```

```
ax.set_xlim(-2, 0.5)
ax.set_ylim(-1.5, 1.5)
ax.set_xlabel(r'$\log([\mathrm{NII}] / \mathrm{H}\alpha)$')
ax.set_ylabel(r'$\log([\mathrm{OIII}] / \mathrm{H}\beta)$')
ax.legend()

plt.tight_layout()
plt.show()
```

```
/Users/isaac/miniconda3/lib/python3.13/site-
packages/pandas/core/arraylike.py:402: RuntimeWarning: invalid value encountered
in log10
  result = getattr(ufunc, method)(*inputs, **kwargs)
```

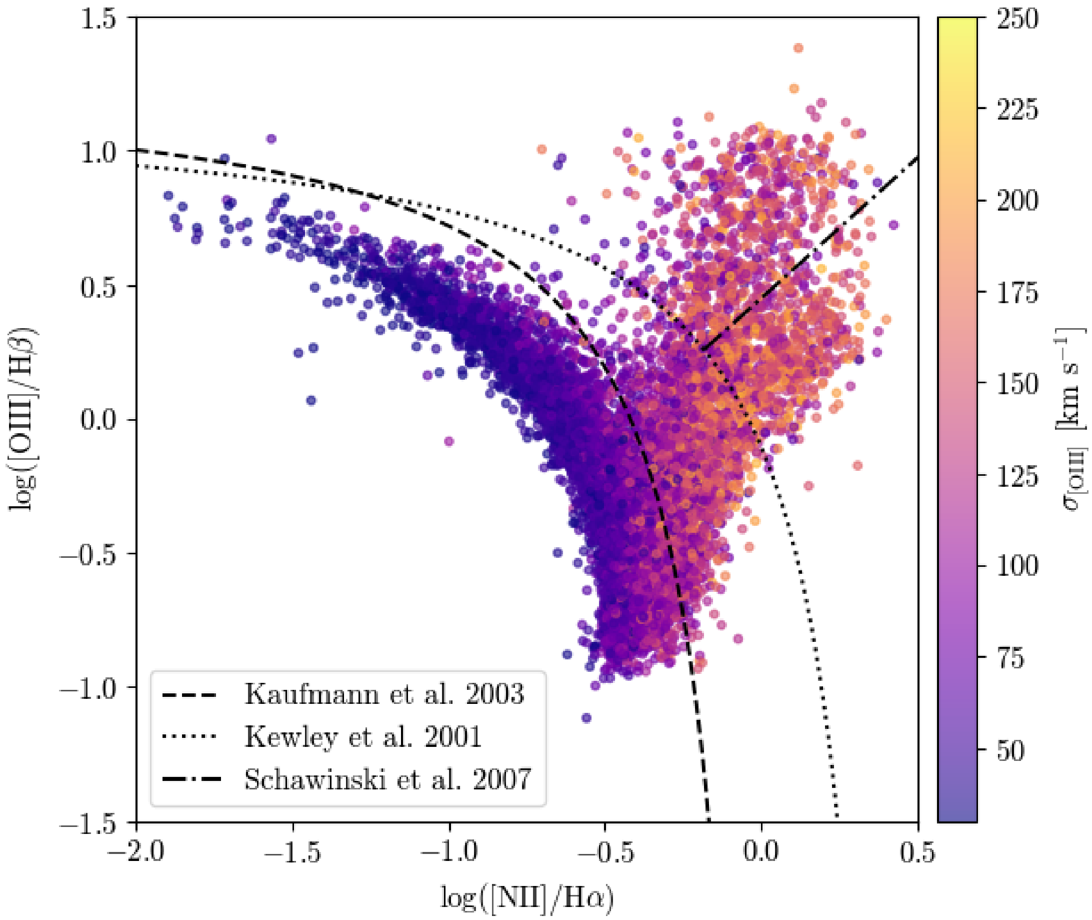


- Higher densities toward the AGN/Seyfert branch indicate denser gas closer to the nucleus
- The star-forming locus, traces lower-density gas in extended H II regions.

3.2 [OIII] Velocity dispersion

[428]:

```
# Plot BPT diagram colored by [OIII] velocity dispersion
fig, ax = plt.subplots(figsize=(6.5, 5.5))

sc = ax.scatter(data['log_nii'], data['log_oiii'],
    c=data['oiii_sigma'],
    cmap="plasma",
    vmin=30, vmax=250,
    s=10,
    alpha=0.6,
)

cbar = fig.colorbar(sc, ax=ax, pad=0.02)
cbar.set_label(r"$\sigma_{[\mathrm{OIII}]}$ [km s$^{-1}$]")

# Classification lines
plt.plot(log_x_kauf, 0.61 / (log_x_kauf - 0.05) + 1.3, color='k',␣
 ↪linestyle='--', label='Kaufmann et al. 2003')
plt.plot(log_x_kew, 0.61 / (log_x_kew - 0.47) + 1.19, color='k', linestyle=':',␣
 ↪label='Kewley et al. 2001')
plt.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45, color='k', linestyle='-.',␣
 ↪label='Schawinski et al. 2007')

ax.set_xlim(-2, 0.5)
ax.set_ylim(-1.5, 1.5)
ax.set_xlabel(r'$\log([\mathrm{NII}]/ \mathrm{H}\alpha)$')
ax.set_ylabel(r'$\log([\mathrm{OIII}]/ \mathrm{H}\beta)$')
ax.legend()

plt.tight_layout()
plt.show()
```



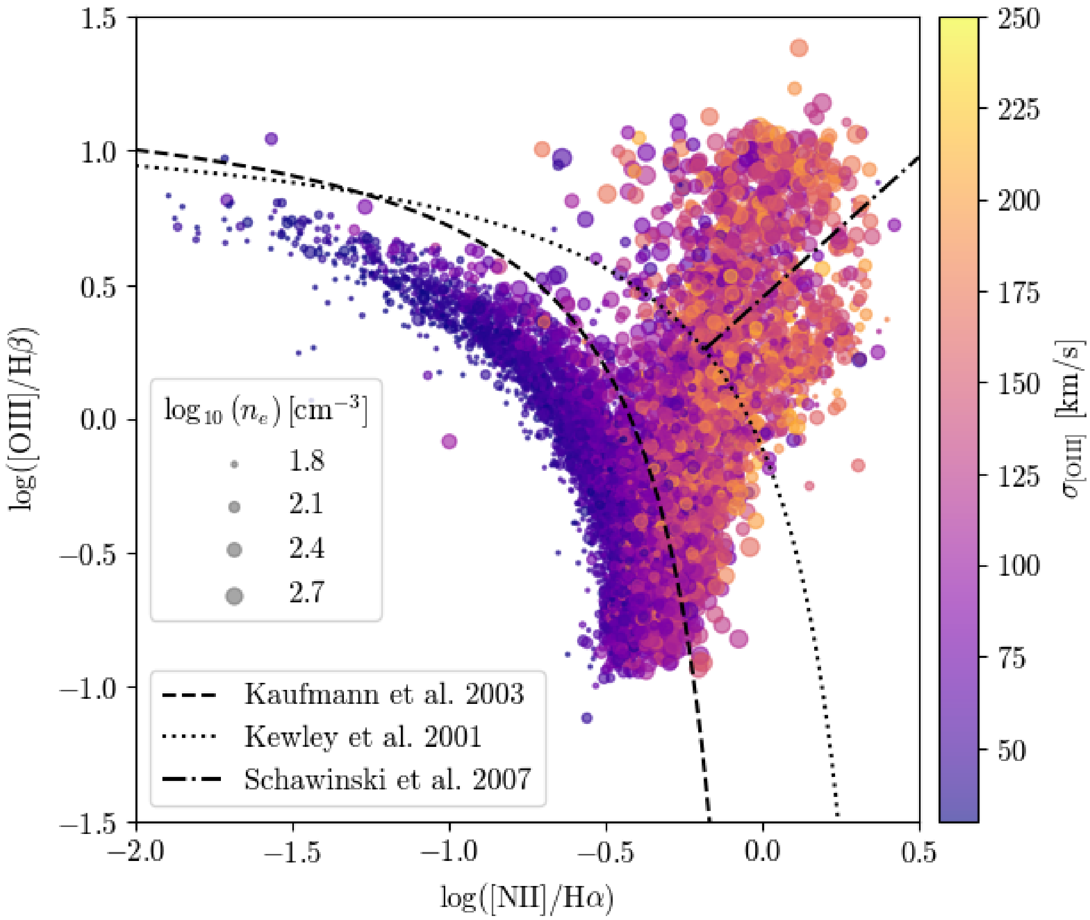
$\sigma_{[\mathrm{OIII}]}$ increases from the star-forming region to the AGN/LINER region. This could reflect:

- deeper potential wells of AGN hosts
- additional non-gravitational broadening (outflows/feedback) of the NLR gas

Mixed plots

[429]:
```
# Plot BPT diagram colored by [OIII] velocity dispersion and sized by electron␣
↪density
fig, ax = plt.subplots(figsize=(6.5, 5.5))

sig = data['oiii_sigma']
logne = np.log10(data['n_e'])

def size_from_logne(lv):
    return np.clip((lv - 1.8) * 40 + 5, 2, None)

sc = ax.scatter(data['log_nii'], data['log_oiii'],
    c=sig, cmap="plasma", vmin=30, vmax=250,
```

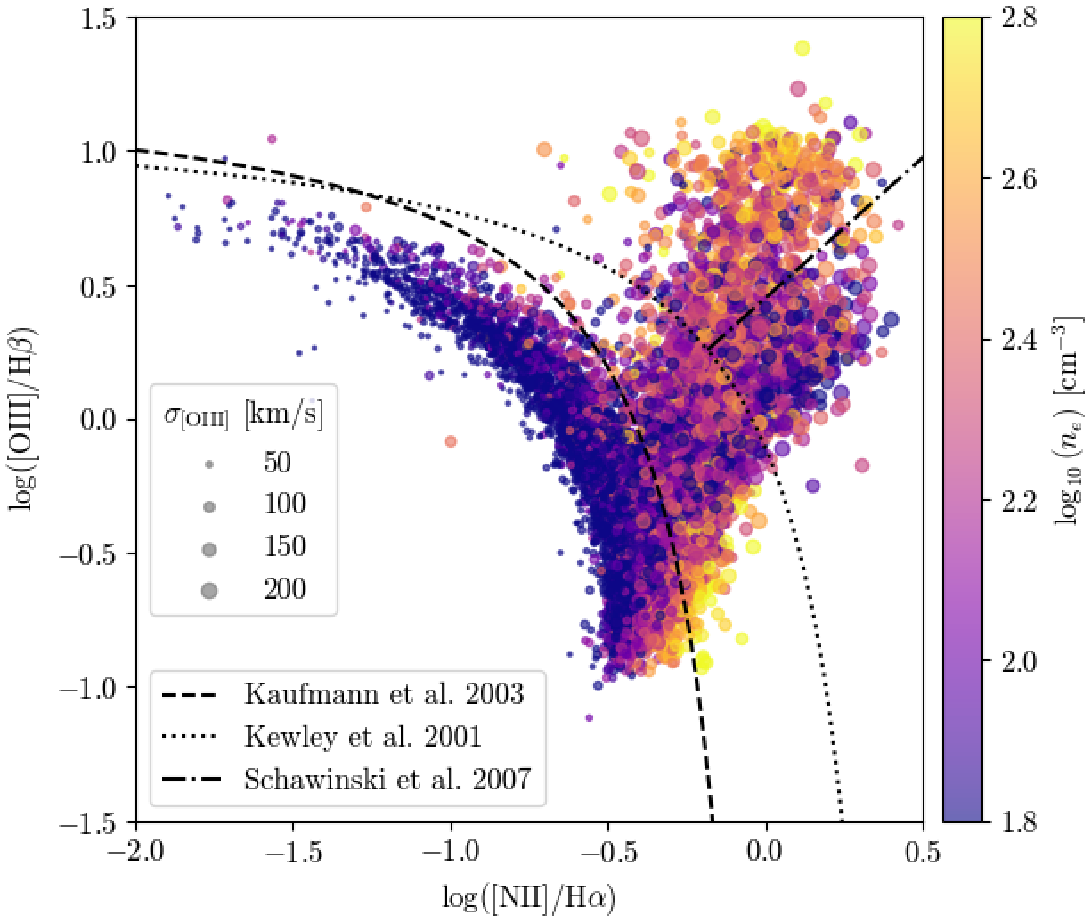
```python
    s=size_from_logne(logne),
    alpha=0.6,
)

cbar = fig.colorbar(sc, ax=ax, pad=0.02)
cbar.set_label(r"$\sigma_{[\mathrm{OIII}]}$ [km/s]")

# Classification lines
plt.plot(log_x_kauf, 0.61 / (log_x_kauf - 0.05) + 1.3, color='k', linestyle='--', label='Kaufmann et al. 2003')
plt.plot(log_x_kew, 0.61 / (log_x_kew - 0.47) + 1.19, color='k', linestyle=':', label='Kewley et al. 2001')
plt.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45, color='k', linestyle='-.', label='Schawinski et al. 2007')


ax.set_xlim(-2, 0.5)
ax.set_ylim(-1.5, 1.5)
ax.set_xlabel(r'$\log([\mathrm{NII}] / \mathrm{H}\alpha)$')
ax.set_ylabel(r'$\log([\mathrm{OIII}]/ \mathrm{H}\beta)$')

# Line legend (upper left)
line_legend = ax.legend(loc='lower left')
ax.add_artist(line_legend)

# Size legend for log_ne
size_levels = [1.8, 2.1, 2.4, 2.7]
size_handles = [
    ax.scatter([], [], s=size_from_logne(lv), color='grey', alpha=0.7,
               label=fr'${lv}$')
    for lv in size_levels
]
ax.legend(handles=size_handles, title=r'$\log_{{10}}(n_e)\,[\mathrm{{cm}}^{{-3}}]$',
          loc='center left',bbox_to_anchor=(0, 0.4))

plt.tight_layout()
plt.show()
```



[430]:

```
# Plot BPT diagram colored by electrond density and sized by [OIII] velocity␣
↪dispersion
fig, ax = plt.subplots(figsize=(6.5, 5.5))

sig = data['oiii_sigma']
logne = np.log10(data['n_e'])

def size_from_sigma(s):
    return np.clip((s - 30) * 0.2 + 2, 2, None)

sc = ax.scatter(data['log_nii'], data['log_oiii'],
    c=logne, cmap="plasma", vmin=1.8, vmax=2.8,
    s=size_from_sigma(sig),
    alpha=0.6,
)

cbar = fig.colorbar(sc, ax=ax, pad=0.02)
```

```python
cbar.set_label(r"$\log_{10}(n_e)$ [cm$^{-3}$]")

# Classification lines
plt.plot(log_x_kauf, 0.61 / (log_x_kauf - 0.05) + 1.3, color='k', linestyle='--', label='Kaufmann et al. 2003')
plt.plot(log_x_kew, 0.61 / (log_x_kew - 0.47) + 1.19, color='k', linestyle=':', label='Kewley et al. 2001')
plt.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45, color='k', linestyle='-.', label='Schawinski et al. 2007')

ax.set_xlim(-2, 0.5)
ax.set_ylim(-1.5, 1.5)
ax.set_xlabel(r'$\log([\mathrm{NII}]/ \mathrm{H}\alpha)$')
ax.set_ylabel(r'$\log([\mathrm{OIII}]/ \mathrm{H}\beta)$')

# Line legend (upper left)
line_legend = ax.legend(loc='lower left')
ax.add_artist(line_legend)

# Size legend for oiii_sigma
sigma_levels = [50, 100, 150, 200]
sigma_handles = [
    ax.scatter([], [], s=size_from_sigma(sv), color='grey', alpha=0.7,
               label=f'{sv}')
    for sv in sigma_levels
]
ax.legend(handles=sigma_handles, title=r'$\sigma_{[\mathrm{OIII}]}$ [km/s]',
          loc='center left', bbox_to_anchor=(0, 0.4))

plt.tight_layout()
plt.show()
```



3.3 SMBH mass

$M_{BH} - \sigma_*$ relation (Tremaine et al. 2002)

$$M_{\rm BH} = 10^{8.13}\left(\frac{\sigma_*}{200{\rm kms}^{-1}}\right)^{4.02} {\rm M}_\odot$$

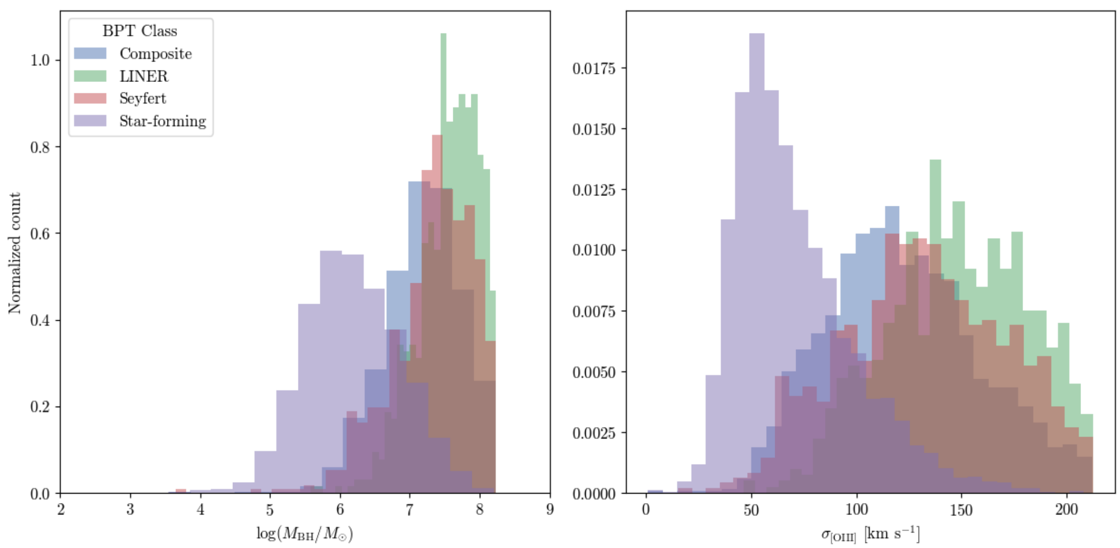
Note that σ is the velocity dispersion of the *host galaxy*. $\sigma_{\rm [OIII]}$ has been used as a proxy of σ_*, but:

- $\sigma_{\rm [OIII]}$ traces NLR gas
- This gas can be broadened by AGN-driven outflows and other non-gravitational motions, biasing the inferred masses high
- This effect strongest precisely in the luminous Seyferts

[431]:
```
def MBH(sigma_oiii):
    return 10**8.13 * (sigma_oiii / 200)**4.02

data['log_MBH'] = np.log10(MBH(data['oiii_sigma']))
```

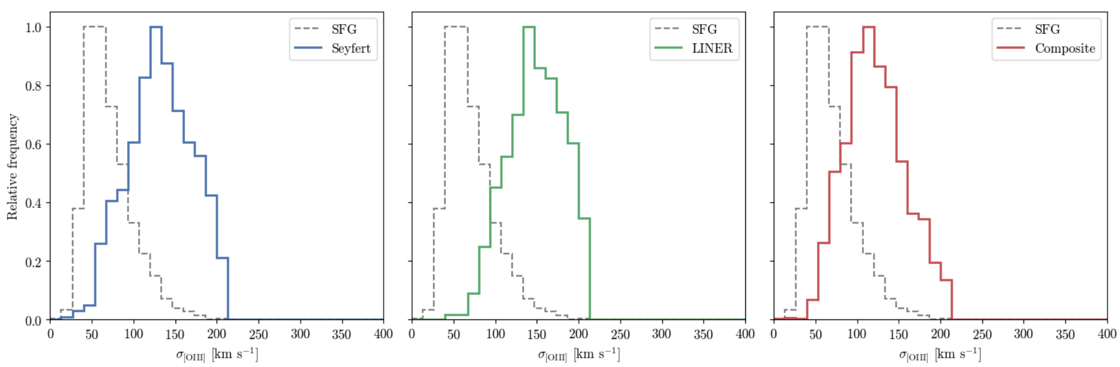
Distribution of BH mass and [OIII] velocity dispersion across BPT classification

[432]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

cols = ['log_MBH', 'oiii_sigma']
labels = [r'$\log(M_{\mathrm{BH}} / M_{\odot})$', r'$\sigma_{[\mathrm{OIII}]}$␣
↪[km s$^{-1}$]']

for label, group in data.groupby('BPT class'):
    axes[0].hist(group[cols[0]], bins=30, alpha=0.5, label=label, density=True)
axes[0].set_xlabel(labels[0])
axes[0].set_xlim(2,9)

for label, group in data.groupby('BPT class'):
    axes[1].hist(group[cols[1]], bins=30, alpha=0.5, label=label, density=True)
axes[1].set_xlabel(labels[1])

axes[0].set_ylabel('Normalized count')
axes[0].legend(title='BPT Class')

plt.tight_layout()
plt.show()
```



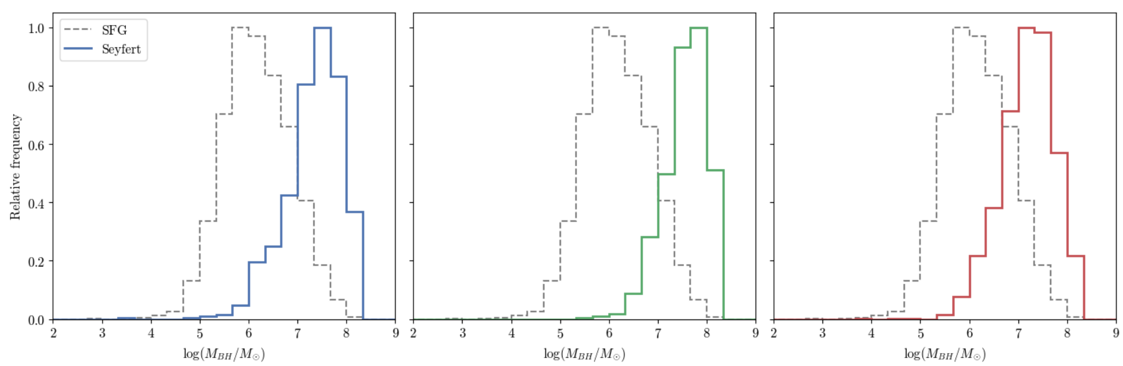
[433]:
```
classes   = ['Seyfert', 'LINER', 'Composite']
colors    = ['C0', 'C1', 'C2']
sfg       = data[data['BPT class'] == 'Star-forming']

fig, axes = plt.subplots(1, 3, figsize=(14.4, 4.8), sharey=True)
```

```
for ax, cls, color in zip(axes, classes, colors):
    grp = data[data['BPT class'] == cls]

    # Compute histograms manually so we can normalize to peak (relative␣
 ↪frequency)
    bins = np.linspace(0, 400, 31)

    sfg_counts, edges = np.histogram(sfg['oiii_sigma'], bins=bins)
    cls_counts, _     = np.histogram(grp['oiii_sigma'], bins=bins)

    sfg_norm = sfg_counts / sfg_counts.max()
    cls_norm = cls_counts / cls_counts.max()

    ax.stairs(sfg_norm, edges, color='grey',  linestyle='--', linewidth=1.5,␣
 ↪label='SFG')
    ax.stairs(cls_norm, edges, color=color,   linestyle='-',  linewidth=2.0,␣
 ↪label=cls)

    ax.set_xlim(0, 400)
    ax.set_ylim(0, 1.05)
    ax.set_xlabel(r'$\sigma_{[\mathrm{OIII}]}$ [km s$^{-1}$]')
    ax.legend()

axes[0].set_ylabel('Relative frequency')

plt.tight_layout()
plt.show()
```



[434]:
```
fig, axes = plt.subplots(1, 3, figsize=(14.4, 4.8), sharey=True)

for ax, cls, color in zip(axes, classes, colors):
    grp = data[data['BPT class'] == cls]
```

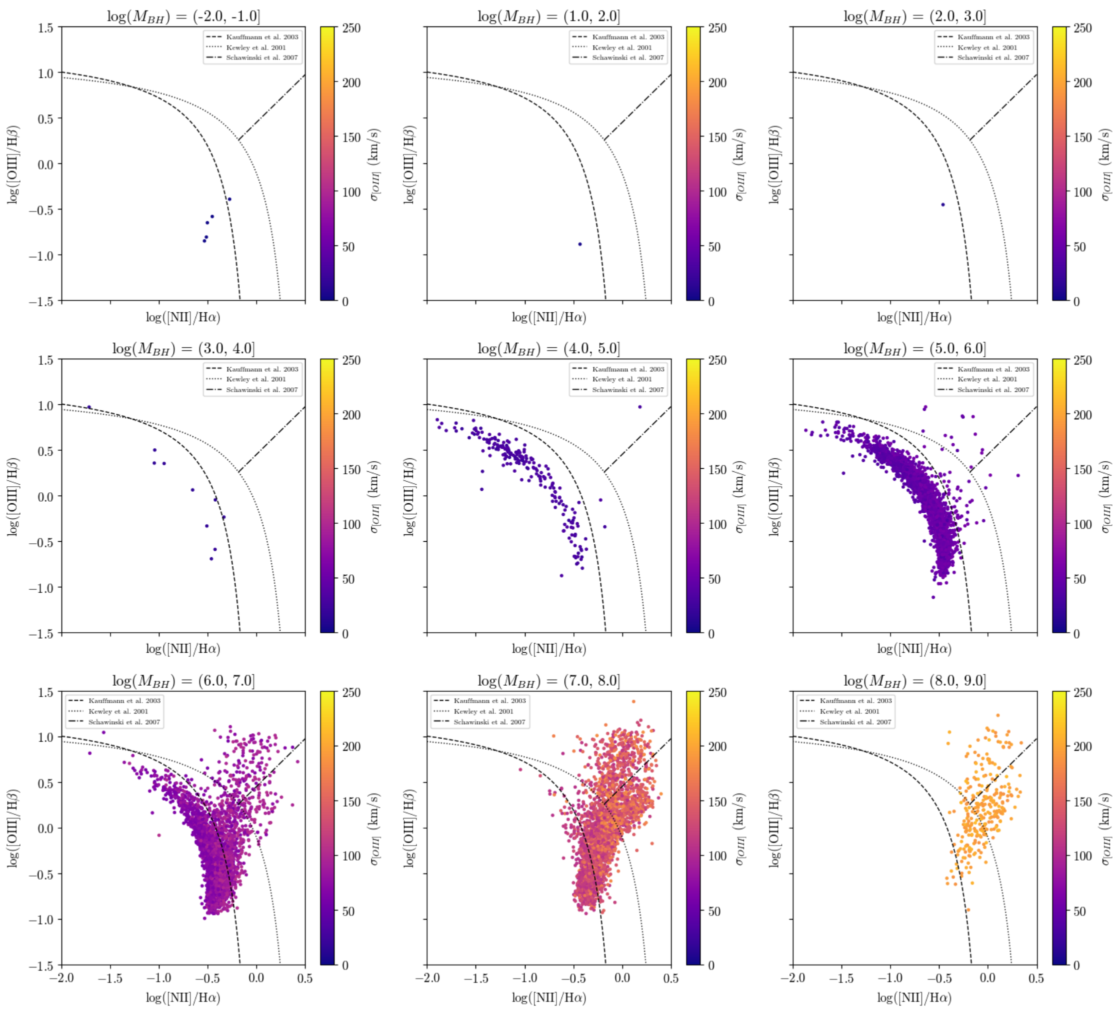
```
    # Compute histograms manually so we can normalize to peak (relative frequency)
    bins = np.linspace(0, 10, 31)

    sfg_counts, edges = np.histogram(sfg['log_MBH'], bins=bins)
    cls_counts, _     = np.histogram(grp['log_MBH'], bins=bins)

    sfg_norm = sfg_counts / sfg_counts.max()
    cls_norm = cls_counts / cls_counts.max()

    ax.stairs(sfg_norm, edges, color='grey',  linestyle='--', linewidth=1.5, label='SFG')
    ax.stairs(cls_norm, edges, color=color,   linestyle='-',  linewidth=2.0, label=cls)

    ax.set_xlim(2, 9)
    #ax.set_ylim(0, 1.05)
    ax.set_xlabel(r'$\log(M_{BH} / M_{\odot})$')

axes[0].set_ylabel('Relative frequency')
axes[0].legend()
plt.tight_layout()
plt.show()
```



[OIII] velocity dispersion across different BH masses

[435]:
```
# Define bins in log(M_BH)
dex = 1
mbh_min = data['log_MBH'].min()
mbh_max = data['log_MBH'].max()
bins = np.arange(np.floor(mbh_min), np.ceil(mbh_max) + dex, dex)

data['mbh_bin'] = pd.cut(data['log_MBH'], bins=bins)
```

```python
groups     = [(lbl, grp) for lbl, grp in data.groupby('mbh_bin')]
n_bins     = len(groups)
ncols      = int(np.ceil(np.sqrt(n_bins)))
nrows      = int(np.ceil(n_bins / ncols))

log_x_kauf  = np.linspace(-2, 0.05, 200)
log_x_kew   = np.linspace(-2, 0.47, 200)
log_x_schaw = np.linspace(np.log10(0.65), 0.5, 200)

log_x_kauf  = log_x_kauf[log_x_kauf < 0.05]
log_x_kew   = log_x_kew[log_x_kew < 0.47]
log_x_schaw = log_x_schaw[log_x_schaw > np.log10(0.65)]

fig, axes = plt.subplots(nrows, ncols, figsize=(5 * ncols, 4.5 * nrows), squeeze=False,sharex='col',sharey='row')
axes_flat = axes.flatten()

for ax, (bin_label, group) in zip(axes_flat, groups):
    sc = ax.scatter(
        np.log10(group['nii_halpha_ratio']),
        np.log10(group['oiii_hbeta_ratio']),
        c=group['oiii_sigma'], cmap='plasma',
        s=5, vmin=0, vmax=250
    )
    fig.colorbar(sc, ax=ax, label=r'$\sigma_{[OIII]}$ (km/s)')
    ax.set_title(fr'$\log(M_{{BH}})$ = {bin_label}')
    ax.set_xlabel(r'$\log(\mathrm{[NII]/H\alpha})$')
    ax.set_ylabel(r'$\log(\mathrm{[OIII]/H\beta})$')

    ax.plot(log_x_kauf,  0.61 / (log_x_kauf  - 0.05) + 1.3,  'k--', lw=1, label='Kauffmann et al. 2003')
    ax.plot(log_x_kew,   0.61 / (log_x_kew   - 0.47) + 1.19, 'k:',  lw=1, label='Kewley et al. 2001')
    ax.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45,           'k-.', lw=1, label='Schawinski et al. 2007')
    ax.set_xlim(-2, 0.5)
    ax.set_ylim(-1.5, 1.5)
    ax.legend(fontsize=7)

for ax in axes_flat[n_bins:]:
    ax.set_visible(False)

plt.tight_layout()
plt.show()
```



[436]:

```python
from matplotlib.animation import FuncAnimation
from IPython.display import HTML

dex = 0.5
mbh_min = data['log_MBH'].min()
mbh_max = data['log_MBH'].max()
bins = np.arange(np.floor(mbh_min), np.ceil(mbh_max) + dex, dex)

data['mbh_bin'] = pd.cut(data['log_MBH'], bins=bins)

groups = [(lbl, grp) for lbl, grp in data.groupby('mbh_bin')]


fig, ax = plt.subplots(figsize=(10, 8))

sc = ax.scatter([], [], c=[], cmap='plasma', vmin=0, vmax=250, s=8)
```

```
cbar = fig.colorbar(sc, ax=ax, label=r'$\sigma_{[OIII]}$ (km/s)')

ax.plot(log_x_kauf,  0.61 / (log_x_kauf  - 0.05) + 1.3,  'k--', lw=1, label='Kauffmann et al. 2003')
ax.plot(log_x_kew,   0.61 / (log_x_kew   - 0.47) + 1.19, 'k:',  lw=1, label='Kewley et al. 2001')
ax.plot(log_x_schaw, 1.05 * log_x_schaw + 0.45,           'k-.', lw=1, label='Schawinski et al. 2007')
ax.set_xlim(-2, 0.5)
ax.set_ylim(-1.5, 1.5)
ax.set_xlabel(r'$\log(\mathrm{[NII]/H\alpha})$')
ax.set_ylabel(r'$\log(\mathrm{[OIII]/H\beta})$')
ax.legend(loc='upper left')
title = ax.set_title('')

def update(frame):
    bin_label, group = groups[frame]
    x = np.log10(group['nii_halpha_ratio'])
    y = np.log10(group['oiii_hbeta_ratio'])
    sc.set_offsets(np.c_[x, y])
    sc.set_array(group['oiii_sigma'].values)
    title.set_text(fr'$\log(M_{{BH}}/M_\odot)$ = {bin_label}  [frame {frame+1}/{len(groups)}]')
    return sc, title

anim = FuncAnimation(fig, update, frames=len(groups), interval=800, blit=True)
plt.close()
HTML(anim.to_jshtml())
```

```
[436]: <IPython.core.display.HTML object>
```

Trends with increasing M_{BH}:

- The BPT location shifts from the star-forming branch (low-mass bins) toward the AGN/Seyfert region (high-mass bins)
- $\sigma_{[OIII]}$ rises across bins (deep purple ~30–60 km/s at low mass → orange ~180–220 km/s at high mass), but is nearly uniform within each panel.
- M_{BH} is computed from $\sigma_{[OIII]}$ via the Tremaine M–σ relation, a monotonic function. Binning by M_{BH} is therefore almost identical to binning by $\sigma_{[OIII]}$.
- This analysis cannot separate gravitational from AGN-driven (outflow) broadening.